A COLLECTION OF FLASHCARDS BY DATA TO VIZ

61 DATAVIZ CAVEATS

The most common charting mistakes, each one next to a better alternative and a short explanation.

YAN HOLTZ · WWW.DATA-TO-VIZ.COM



THE CHECKLIST

Run through it before you publish a chart. Page numbers point to the card.

MISLEADING CHARTS

- [ ] Do not cut the y-axis 3
- [ ] Boxplots hide information 3
- [ ] Avoid counterintuitive visuals 3
- [ ] Error bars on barplots 4
- [ ] Dual y-axis 4
- [ ] Avoid 3D 4
- [ ] Avoid cherry-picking data 5
- [ ] Draw proportions accurately 5
- [ ] Respect time intervals 5
- [ ] Show the sample size 6
- [ ] Beware of Simpson's paradox 6

WRONG CHART TYPE

- [ ] Avoid the moiré effect with lollipops 6
- [ ] What's wrong with pie charts? 7
- [ ] Radial bar charts distort perception 7
- [ ] Maybe you don't need a map 7
- [ ] Two groups? Try a dumbbell plot 8
- [ ] Use circular barplots with care 8

COLOR

- [ ] Use color intentionally 8
- [ ] Optimize background colors 9
- [ ] Color contrast 9
- [ ] Avoid pure, flashy colors 9
- [ ] Use a sequential palette for numbers 10
- [ ] Too many colors kill the story 10
- [ ] Don't use a gradient for groups 10
- [ ] Avoid the rainbow palette 11
- [ ] Diverging data, diverging palette 11
- [ ] Think of colorblind readers 11

DESIGN

- [ ] Align everything with something 12
- [ ] Two fonts are enough 12
- [ ] Do you really need that box? 12
- [ ] Font size matters 13
- [ ] Group what belongs together 13
- [ ] Make the title say something 13
- [ ] Declutter: less is more 14

MAPS

- [ ] Don't move countries around 14
- [ ] Beware of the Mercator projection 14
- [ ] Normalize your maps 15
- [ ] Show missing data as missing 15
- [ ] Don't let small regions disappear 15

LABELS AND LEGENDS

- [ ] Prefer horizontal labels 16
- [ ] Minimize the use of legends 16
- [ ] Don't put labels on top of the data 16
- [ ] Round your numbers 17
- [ ] Annotation is crucial 17
- [ ] Don't label every tick 17

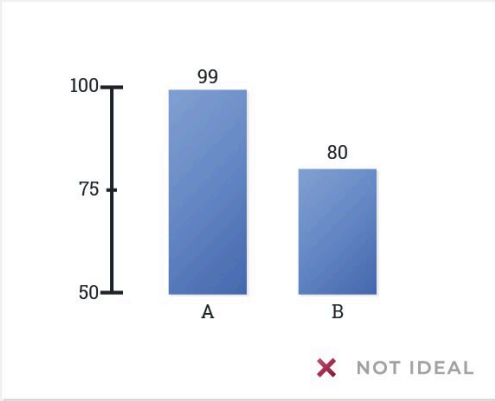
DASHBOARDS AND DATA STORIES

- [ ] Keep colors consistent across charts 18
- [ ] Share the y-axis to compare charts 18
- [ ] Use margins between charts 18
- [ ] Link charts that share an axis 19

OTHER

- [ ] Rank your barplot 19
- [ ] Histograms: mind the bin size 19
- [ ] Size bubbles by area, not radius 20
- [ ] The spaghetti chart 20
- [ ] Connect the dots 20
- [ ] Z-index matters 21
- [ ] UpSet plots 21
- [ ] Stacked charts are hard to read 21
- [ ] Mind your scatter plot's shape 22
- [ ] Make date axes easy to follow 22
- [ ] Reorder your heatmap 22
- [ ] Beware of overplotting 23

DO NOT CUT THE Y AXIS



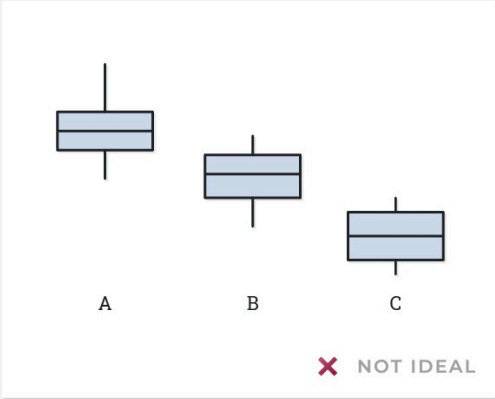
Avoid cutting the Y axis when comparing groups, as it can distort differences and mislead the interpretation of the data.



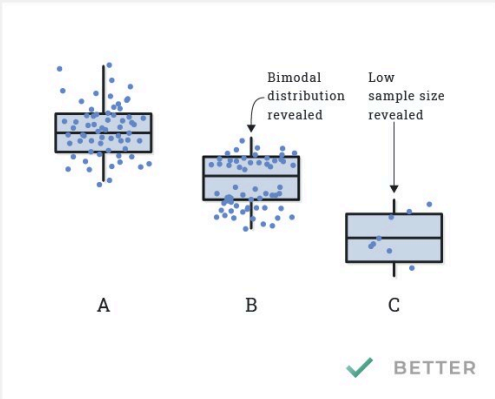
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BOXPLOTS HIDE INFORMATION



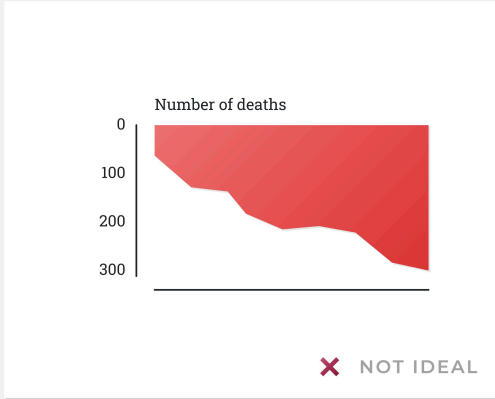
A boxplot simplifies data but can **hide important details**, such as the sample size and distribution patterns. Consider adding individual data points or opting for a different chart type, such as a violin or raincloud plot.



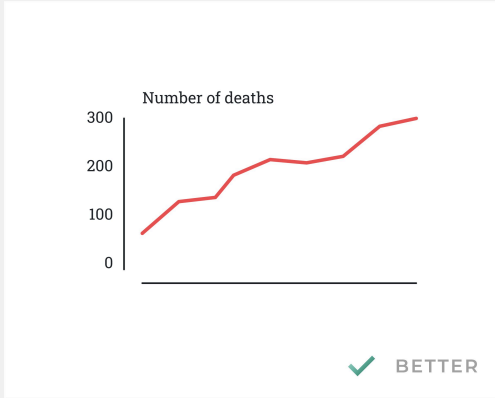
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AVOID COUNTERINTUITIVE VISUALS



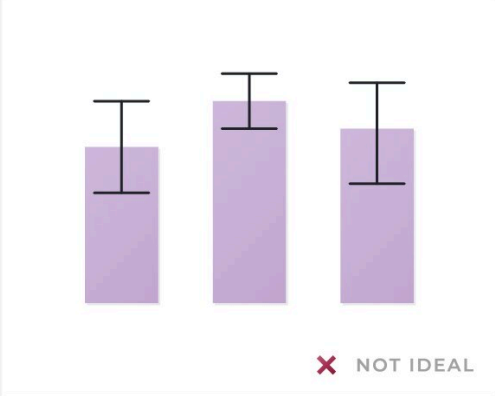
Visuals should follow common conventions to avoid confusion. Misleading color choices, reversed timelines, or inverted Y-axes make it harder to understand the data.



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ERROR BARS ON BARPLOT



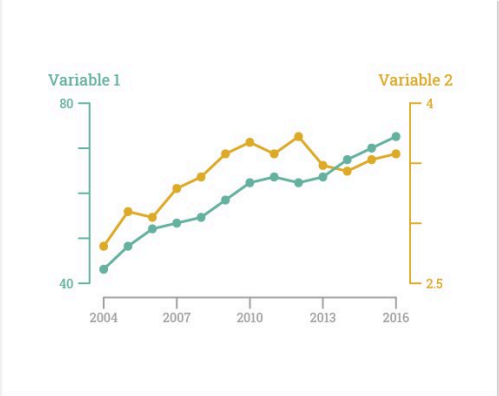
Error bars obscure the full distribution. If individual data points are available, consider a chart that displays them instead.
When using error bars, always clarify whether they represent a confidence interval, standard error, or standard deviation.



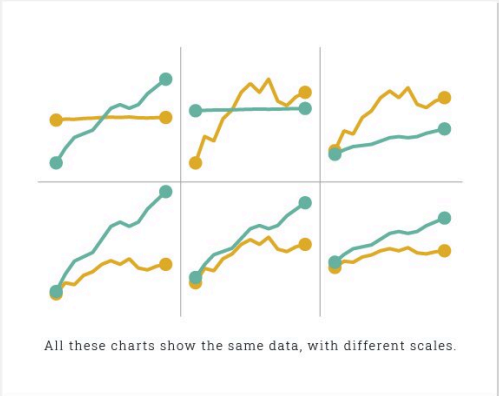
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DUAL Y AXIS



Dual Y-axis charts can be misleading if the scales are manipulated to exaggerate correlations. Instead, consider using two separate charts or normalizing the data for clearer comparisons.

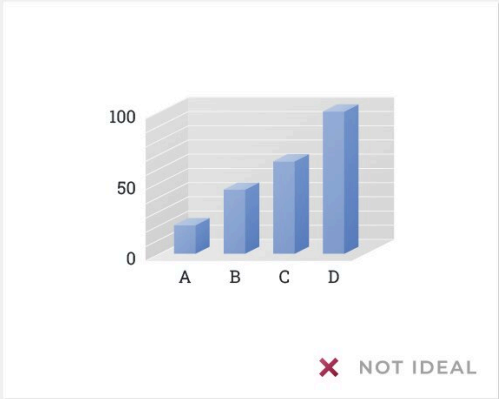


All these charts show the same data, with different scales.

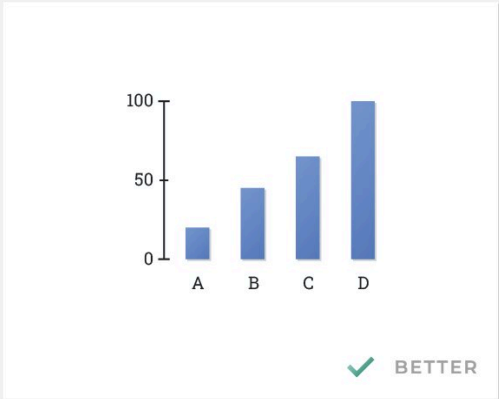
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AVOID 3D



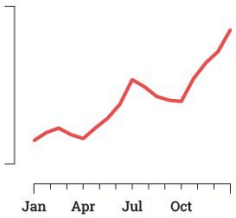
3D distorts perception without adding value. Stick to 2D.



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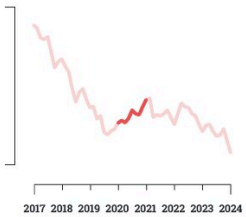
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AVOID CHERRY-PICKING DATA



✕ NOT IDEAL

Cherry-picking data by zooming in on a specific section can mislead viewers. Always show the broader context to reveal the true trend.

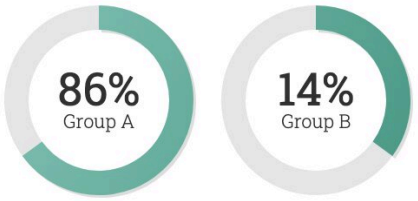


✓ BETTER

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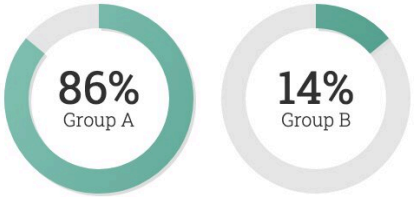
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DRAW PROPORTIONS ACCURATELY



✕ NOT IDEAL

When shapes don't match the numbers, readers trust the shapes and get the wrong picture. Always draw arcs, bars and areas from the data, never by eye.

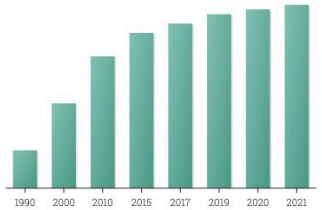


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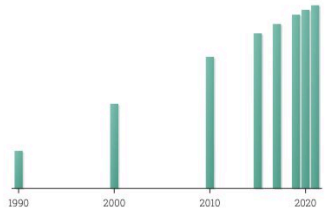
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RESPECT TIME INTERVALS



✕ NOT IDEAL

Evenly spaced bars for unevenly spaced dates distort the trend: here, steady growth looks like a slowdown. Put time on a true scale, so that gaps between dates show as gaps on the chart.

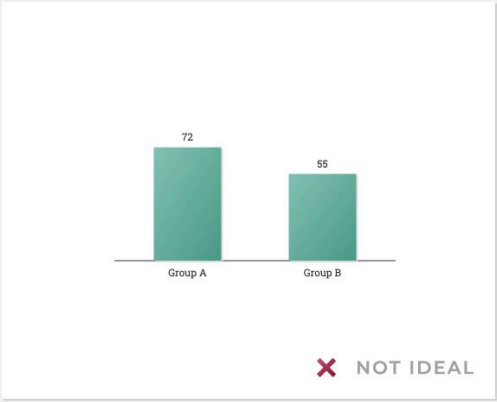


✓ BETTER

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SHOW THE SAMPLE SIZE



A mean alone hides how much data stands behind it: 8 values or 1,000 look exactly the same. Show the individual points, or at least the sample size, so readers can judge how solid it is.



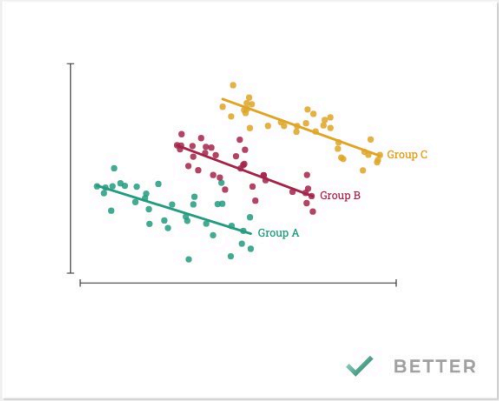
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BEWARE OF SIMPSON'S PARADOX



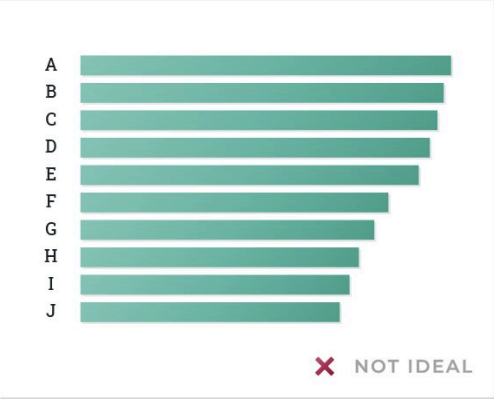
Overall, x and y seem to rise together. Split the data by group, and the trend reverses in each one. Before trusting a correlation, check whether a hidden variable drives it.



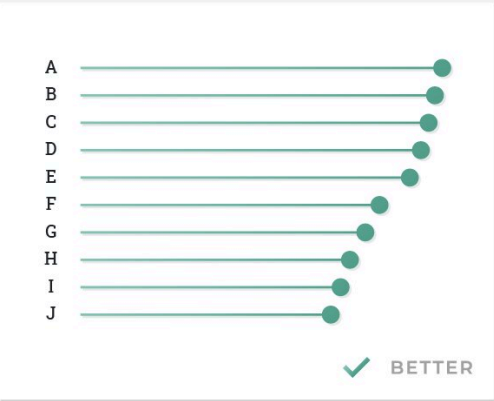
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AVOID THE MOIRÉ EFFECT WITH LOLLIPOPS



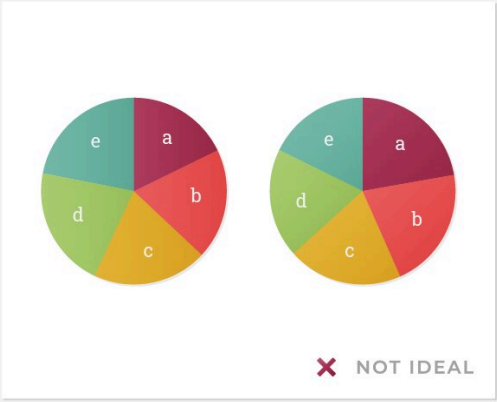
When a barplot contains many long bars, a Moiré effect can occur, creating an unwanted optical illusion. Switching to a lollipop chart reduces visual clutter and prevents this illusion and makes the chart cleaner.



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WHAT'S WRONG WITH PIE CHARTS?



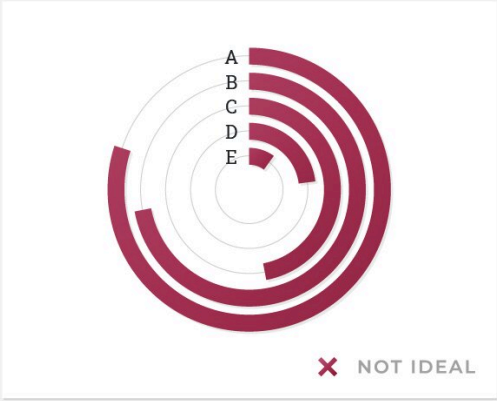
The human eye struggles to accurately read angles, making pie charts less effective for comparisons. Opt for chart types that make it easier to compare sizes. Pie charts are acceptable for very simple scenarios.



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RADIAL BAR CHARTS: DISTORTING PERCEPTION



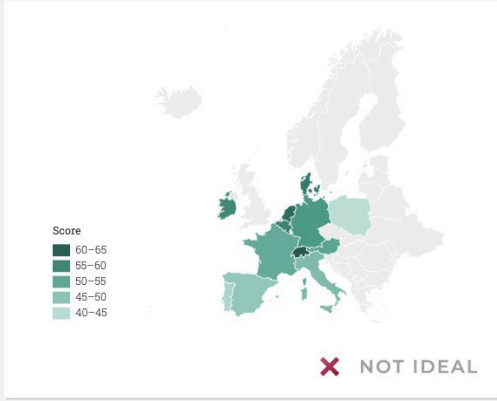
Radial bar charts with bars wrapping around the clock can mislead, as outer bars appear larger due to the layout, not the data. Use a standard bar chart to avoid this distortion.



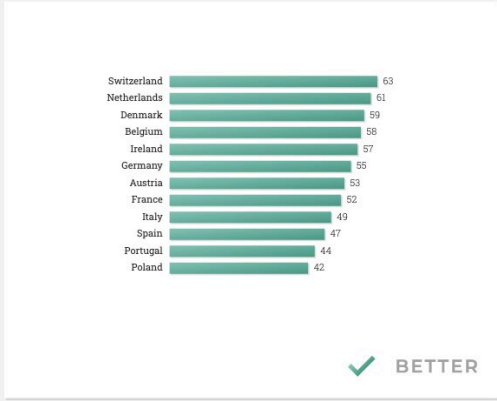
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MAYBE YOU DON'T NEED A MAP



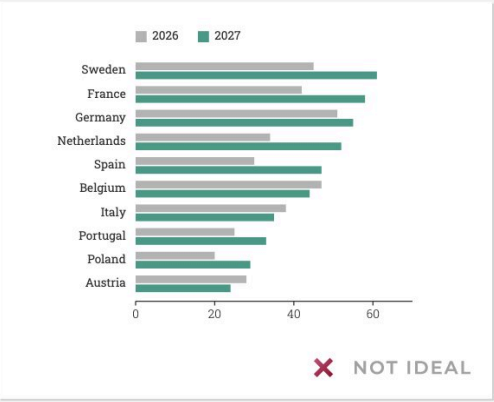
If the question is "who ranks first?", colored shapes make readers squint and guess. When geography is not the point, a sorted bar chart compares values far more precisely.



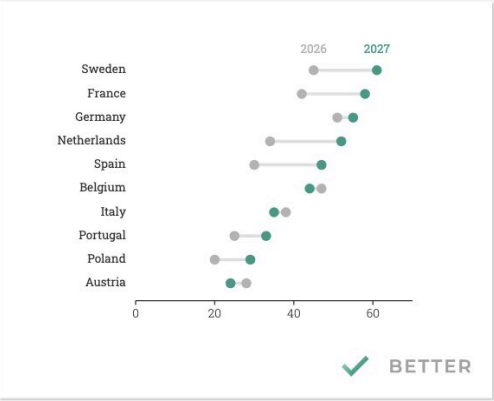
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TWO GROUPS? TRY A DUMBBELL PLOT



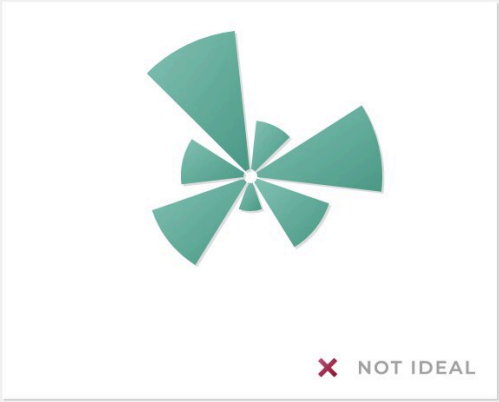
Side-by-side bars double the ink and leave readers to estimate the gap between them on their own. When the change is the story, a dumbbell plot shows it directly, and uses half the ink.



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USE CIRCULAR BARPLOTS WITH CARE



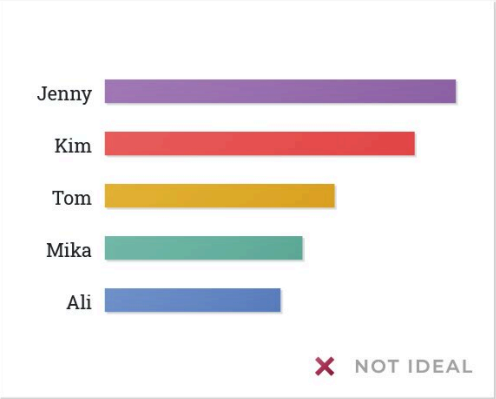
Bars get wider away from the center, so large values look even larger than they are. If you go circular, use many bars, a clear pattern, and a large inner circle.



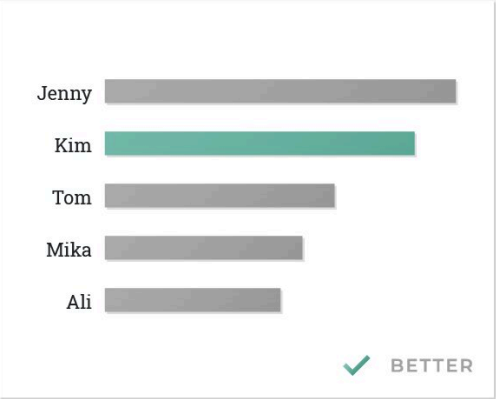
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USE COLOR INTENTIONALLY



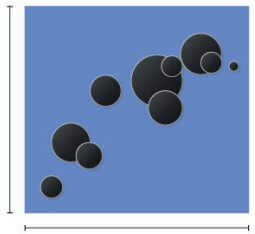
Use color only when it adds meaning to your chart. Avoid assigning different colors to each element in a graph unless they signify distinct categories. A consistent color scheme keeps the visualization clean and minimizes distraction.



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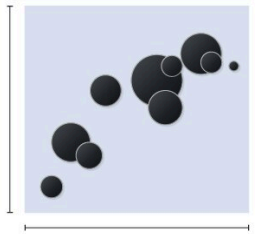
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OPTIMIZING BACKGROUND COLORS



✕ NOT IDEAL

Use high-luminance, low-saturation backgrounds in light mode, and softened dark tones instead of pure black in dark mode.

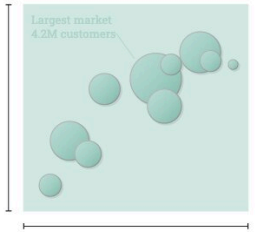


✓ BETTER

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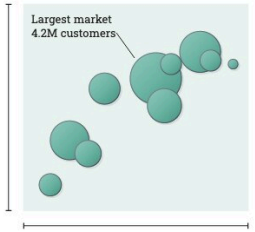
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COLOR CONTRAST



✕ NOT IDEAL

Low contrast makes chart elements hard to tell apart, and text hard to read. Keep enough contrast between marks, background and labels, and write annotations in a dark color.

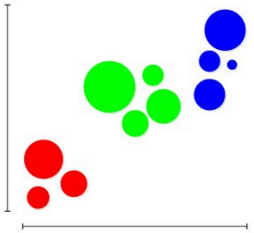


✓ BETTER

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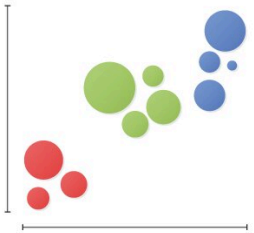
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AVOID PURE, FLASHY COLORS



✕ NOT IDEAL

Pure red, green, and blue are too harsh on the eyes, create contrast issues, and are problematic for colorblind users. Use softer, slightly desaturated colors for better readability.

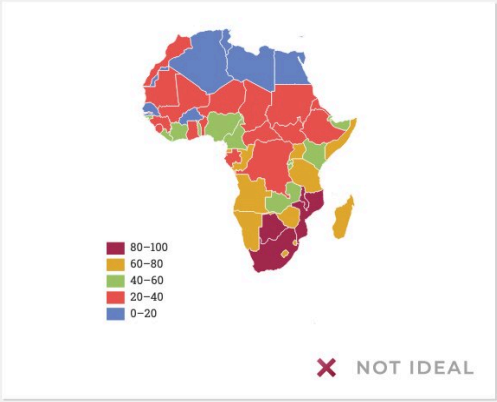


✓ BETTER

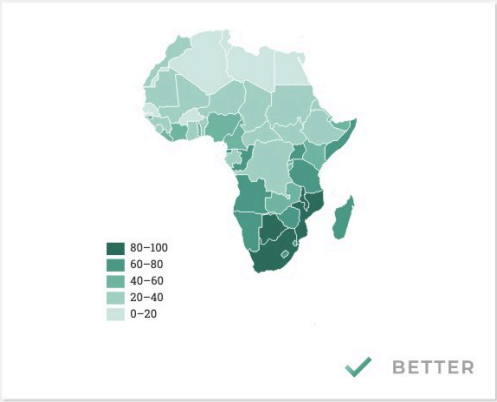
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USE A SEQUENTIAL PALETTE FOR NUMBERS



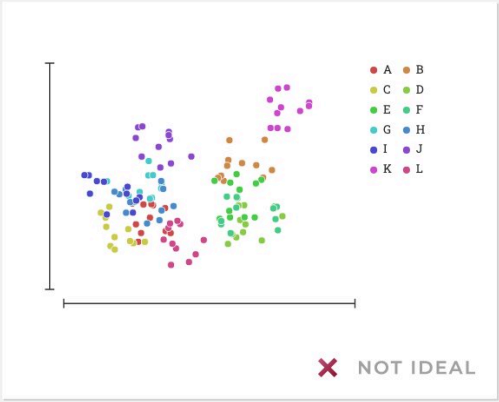
Categorical colors have no natural order: readers can't tell which color means more without constantly checking the legend. For numeric values, use a sequential palette from light to dark.



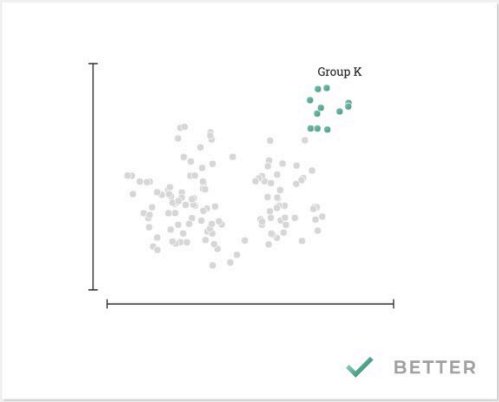
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TOO MANY COLORS KILL THE STORY



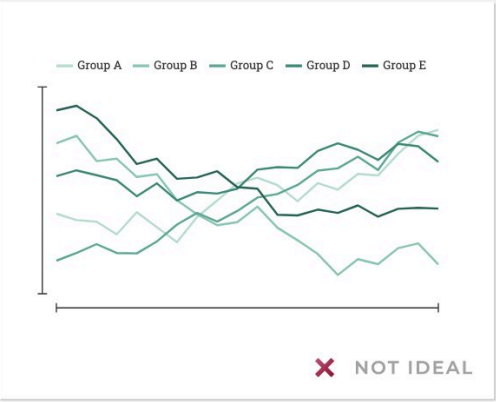
Beyond a handful of colors, readers can no longer match them to the legend. Find the story in your data, highlight only what matters, and leave the rest in grey.



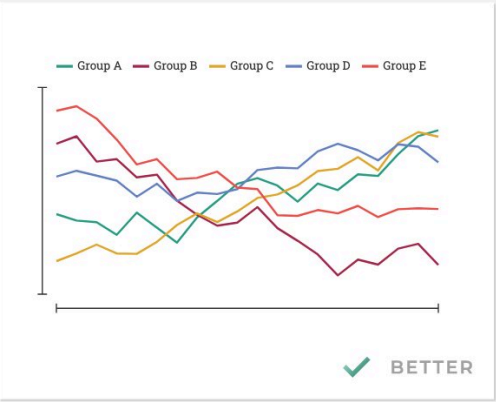
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DON'T USE A GRADIENT FOR GROUPS



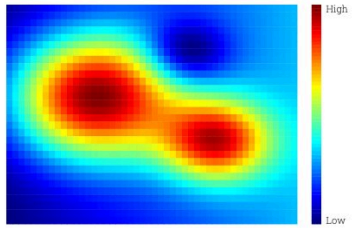
Shades of a single color are hard to tell apart and suggest an order between groups that doesn't exist. For categories, use clearly distinct hues, and keep gradients for numeric values.



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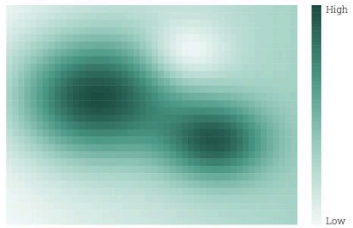
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AVOID THE RAINBOW PALETTE



✕ NOT IDEAL

Rainbow palettes draw sharp bands that aren't in the data, and their bright yellow grabs attention for no reason. Use a perceptually uniform palette that goes smoothly from light to dark.

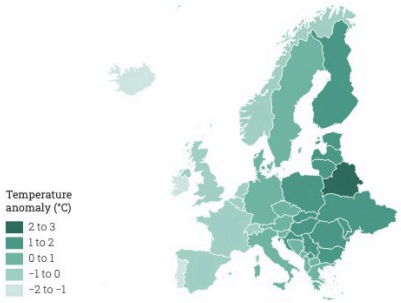


✓ BETTER

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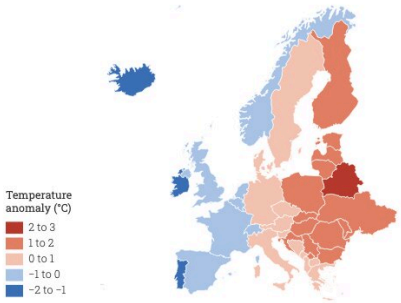
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DIVERGING DATA, DIVERGING PALETTE



✕ NOT IDEAL

When values fall on both sides of a meaningful midpoint, like zero, a sequential palette hides which side they are on. Use a diverging palette: one hue for each direction, meeting at a light neutral color at the midpoint.

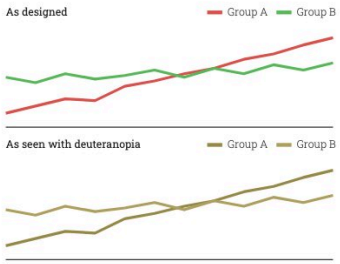


✓ BETTER

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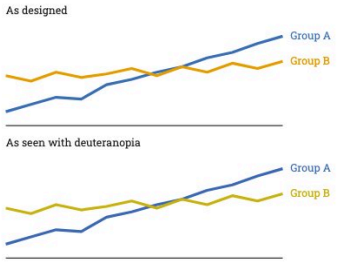
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THINK OF COLORBLIND READERS



✕ NOT IDEAL

About 1 man in 12 has a color vision deficiency, and red versus green is the classic trap. Pick colorblind-safe pairs such as blue and orange, and label lines directly whenever you can.

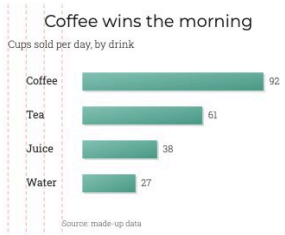


✓ BETTER

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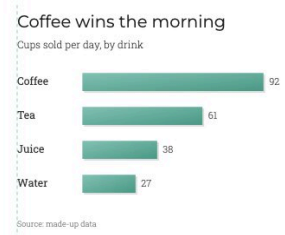
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ALIGN EVERYTHING WITH SOMETHING



✕ NOT IDEAL

When the title, labels and notes each start somewhere different, the chart looks messy and is harder to scan. Snap every element to a shared edge: the fewer alignment lines, the calmer the figure.

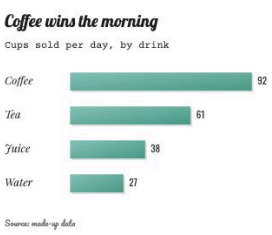


✓ BETTER

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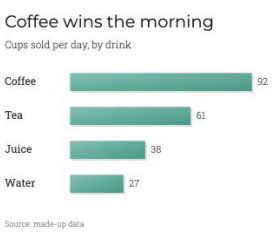
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TWO FONTS ARE ENOUGH



✕ NOT IDEAL

Every extra typeface adds visual noise and makes a chart look improvised. Stick to two fonts at most, and use size and weight to build the hierarchy.

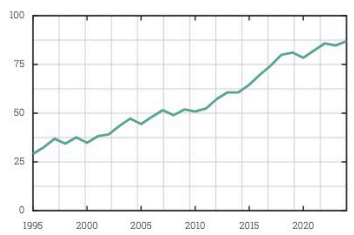


✓ BETTER

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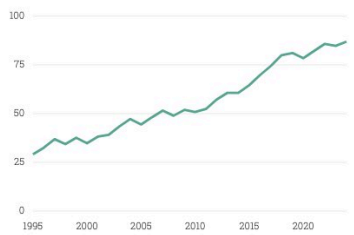
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DO YOU REALLY NEED THAT BOX?



✕ NOT IDEAL

Many R and Python defaults wrap every chart in a box, with ticks on all sides and a dense grid. It is pure clutter: drop the frame, keep a few light gridlines, and let the data stand out.

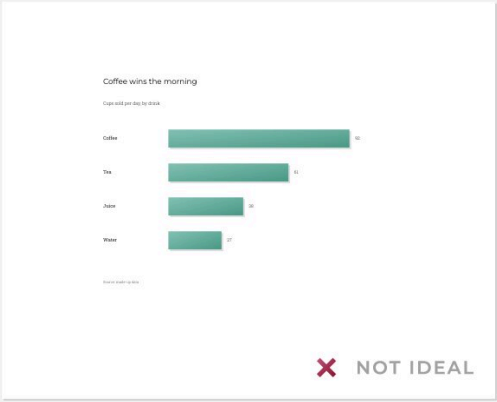


✓ BETTER

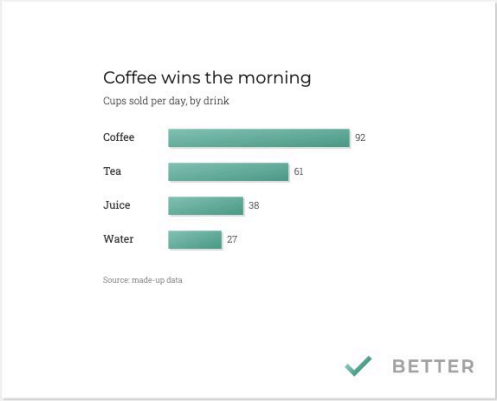
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FONT SIZE MATTERS



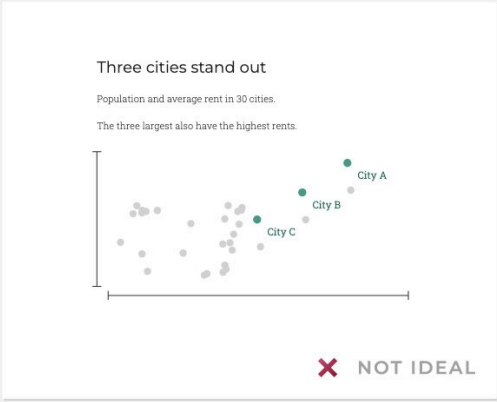
Tiny text forces readers to squint, and becomes unreadable on slides and phones. Never go below 14px for anything meant to be read, and make the title bigger still.



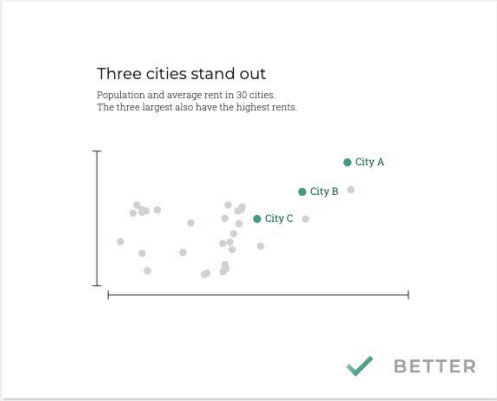
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GROUP WHAT BELONGS TOGETHER



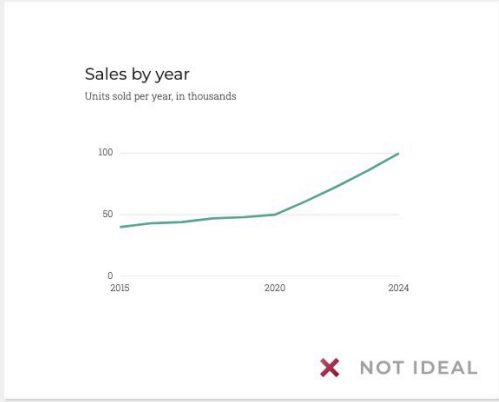
Spacing tells readers what goes together. Loose line height and floating labels blur the structure. Keep related items tight, put labels right next to their data, and leave more room between groups.



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MAKE THE TITLE SAY SOMETHING



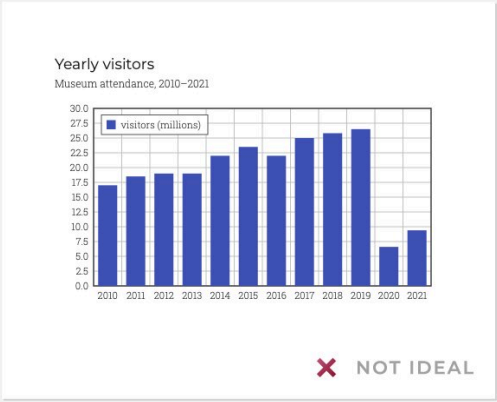
A generic title like "Sales by year" leaves readers to find the message on their own. Write the takeaway in the title, and highlight it on the chart.



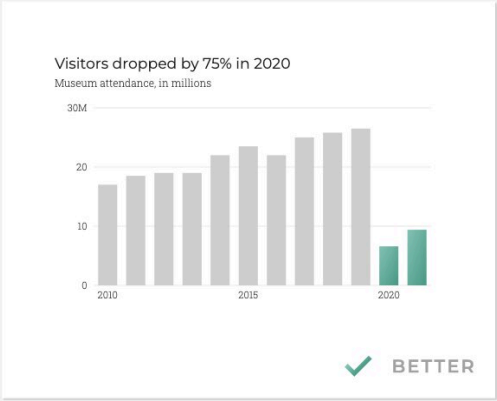
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DECLUTTER: LESS IS MORE



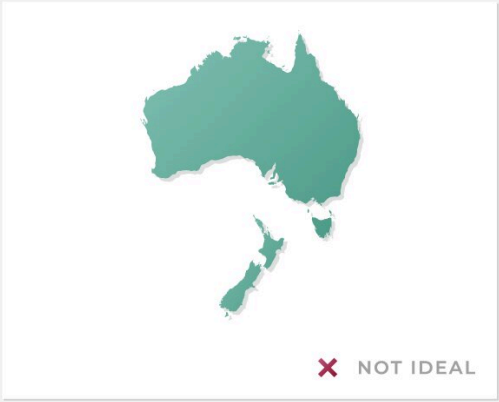
Frames, dense grids, legends and decimals all compete with your message for the reader's eye. Remove what doesn't serve your point, and let the title and colors carry it.



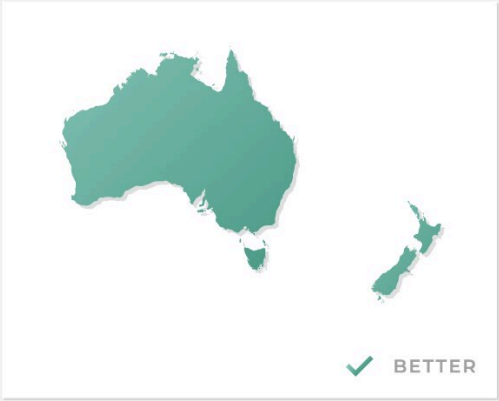
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DON'T MOVE COUNTRIES AROUND



Moving a country to save space, or dropping it because it doesn't fit, distorts geography and can offend its people. Keep every country where it belongs, even if it costs some room.



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BEWARE OF THE MERCATOR PROJECTION



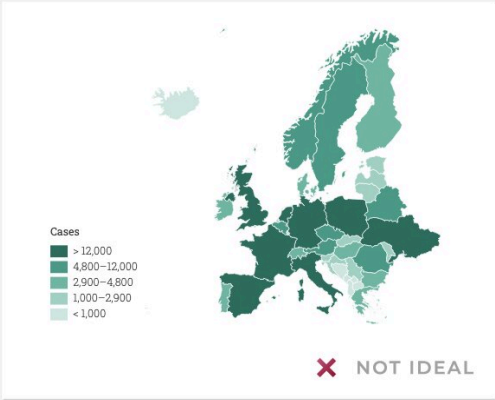
Mercator inflates areas far from the equator: Greenland looks as big as Africa, which is actually 14 times larger. To compare areas, use an equal-area projection like Equal Earth.



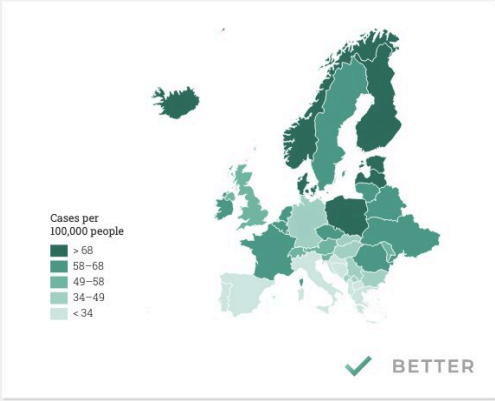
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NORMALIZE YOUR MAPS



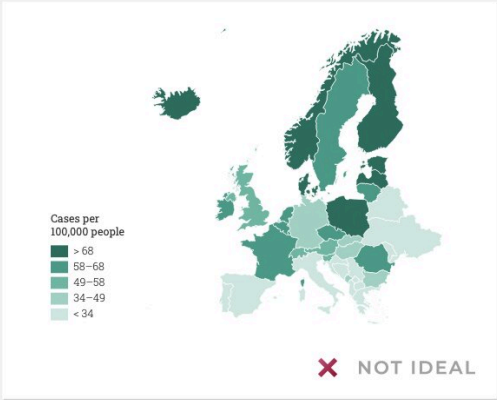
A map of raw counts is mostly a map of where people live: the biggest countries always look the worst. Divide by population, or by area, so that colors show a rate rather than a size.



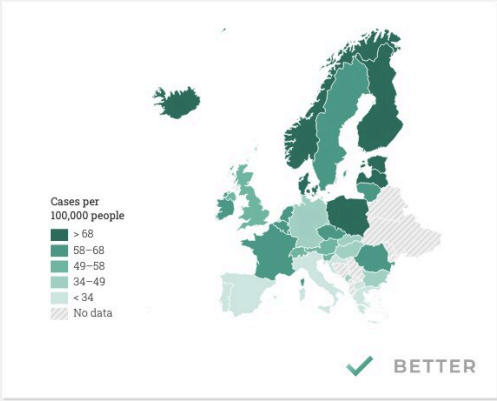
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SHOW MISSING DATA AS MISSING



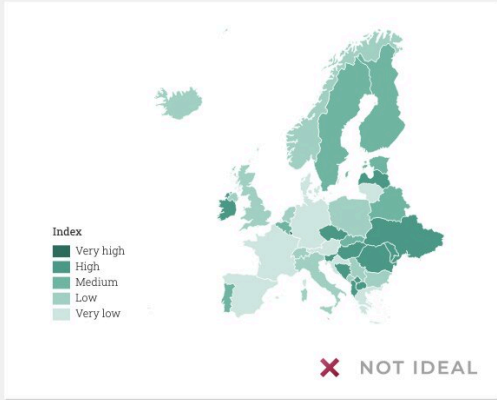
When countries without data get the lightest color, readers take them for low values. Give missing data its own neutral style, like grey hatching, and name it in the legend.



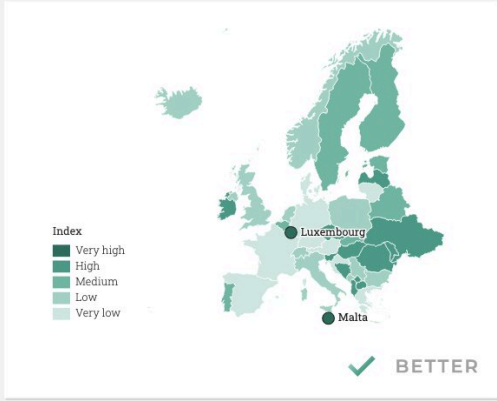
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DON'T LET SMALL REGIONS DISAPPEAR



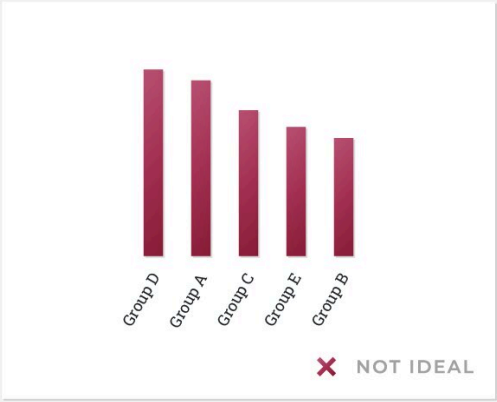
On a choropleth, small countries are only a few pixels wide: here, the two highest values are invisible. Call them out with markers and labels, or use a tile map where every region gets the same size.



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PREFER HORIZONTAL LABELS

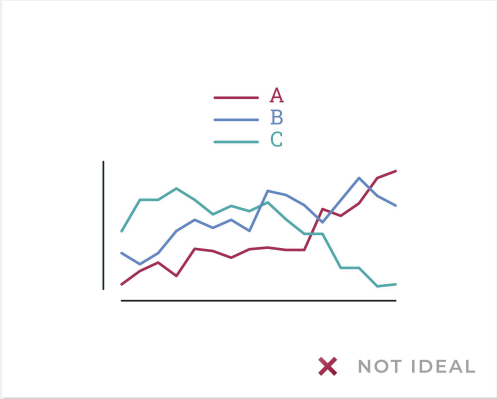


Always aim for a chart type that makes your labels readable. For instance, a horizontal barplot will often be cleaner than a vertical version.

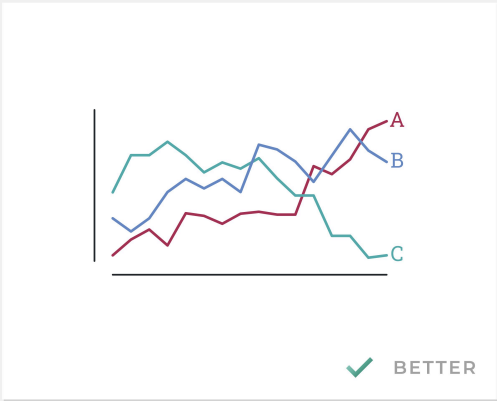


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MINIMIZE THE USE OF LEGENDS



Reading a legend requires significant effort, as the reader must constantly move their eyes back and forth to match labels with graph elements. Whenever possible, opt for direct labeling instead.



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DON'T PUT LABELS ON TOP OF THE DATA

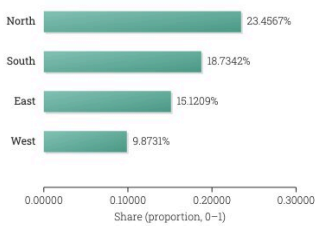


Labels and annotations drawn over lines or shapes hide the data and are hard to read. Place them in empty space, right next to what they describe, such as at the end of a line.



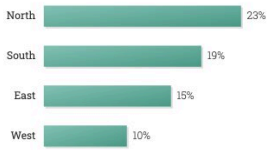
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ROUND YOUR NUMBERS



✕ NOT IDEAL

Too many decimals add noise, and suggest a precision the data doesn't have. Round to what matters, and drop axes that only repeat the labels.

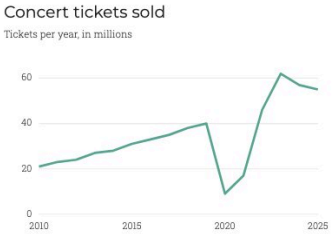


✓ BETTER

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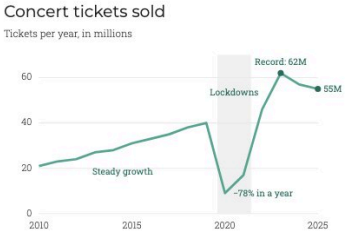
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ANNOTATION IS CRUCIAL



✕ NOT IDEAL

A bare chart leaves readers guessing, and you only have a few seconds to hold their attention. Add short notes, shaded areas and value labels that point straight at what matters.

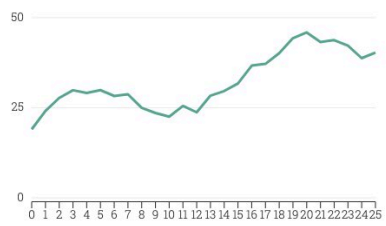


✓ BETTER

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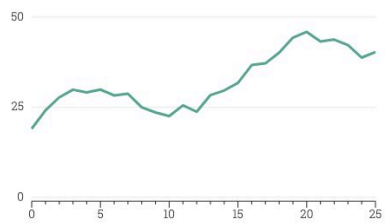
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DON'T LABEL EVERY TICK



✕ NOT IDEAL

Labelling every tick crowds the axis, and on small screens the numbers soon touch or overlap. Label round steps only, and keep shorter ticks in between to show the scale.

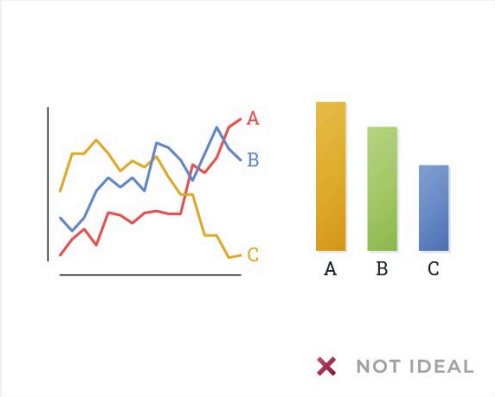


✓ BETTER

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KEEP COLORS CONSISTENT ACROSS CHARTS



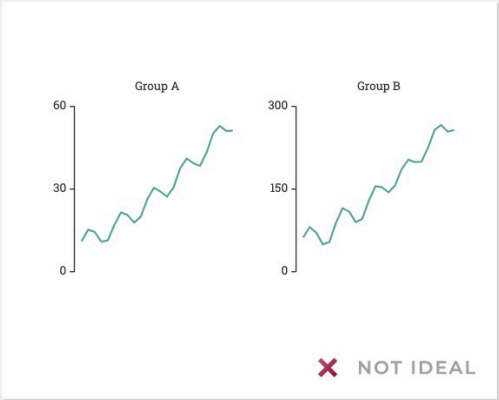
Use consistent colors for the same groups across charts to enhance clarity. Changing colors can confuse the viewer, while a uniform scheme keeps comparisons clear and intuitive.



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SHARE THE Y AXIS TO COMPARE CHARTS



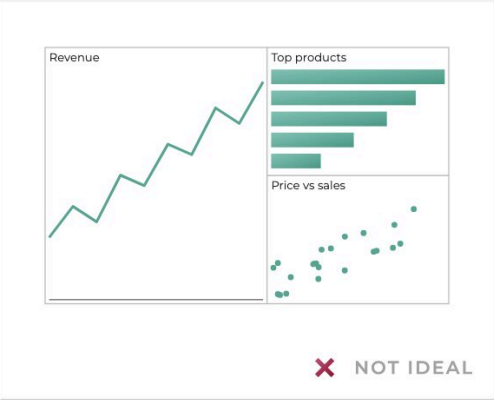
When charts sit side by side to compare the same measure, different y scales make them look alike. Use the same y axis on all of them so that heights can be compared directly.



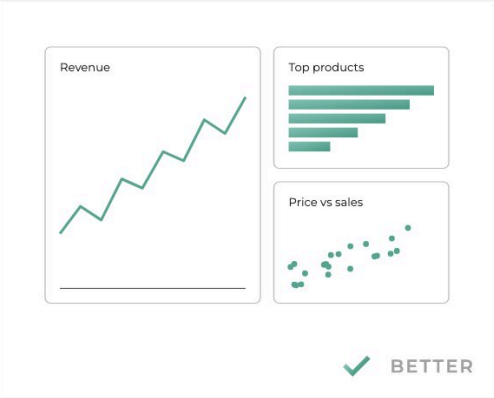
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USE MARGINS BETWEEN CHARTS



When charts touch each other, the page feels crowded and it's hard to tell where one chart ends. Leave clear gaps between them, and some padding inside each one.



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LINK CHARTS THAT SHARE AN AXIS



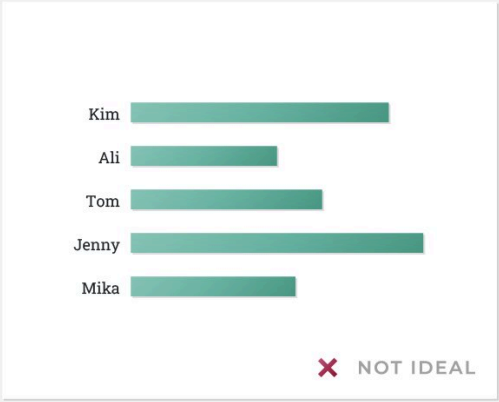
When charts share dates or groups, a tooltip on just one of them leaves readers hunting for the match. On hover, highlight the same date or group on every chart at once.



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RANK YOUR BARPLOT



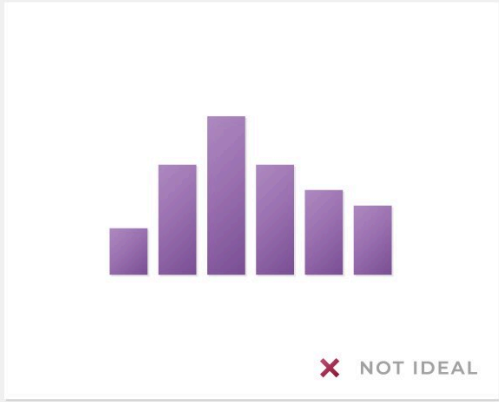
When displaying several items, ordering them makes the graph much more insightful: the ranking can be grasped instantly, with no unnecessary cognitive effort.



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HISTOGRAM? MIND THE BIN SIZE



Always adjust the bin size of a histogram, as different binning can uncover hidden patterns or provide a clearer understanding of the data.

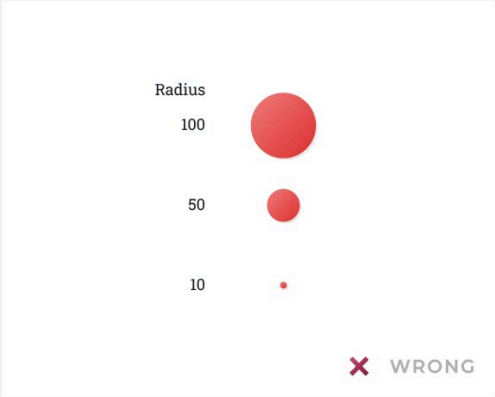


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52 OTHER

MAP BUBBLE SIZE BY AREA, NOT RADIUS



Mapping data to the radius of a bubble creates a visual distortion because the area grows with the square of the radius. This makes the bubbles appear much larger than the actual values.

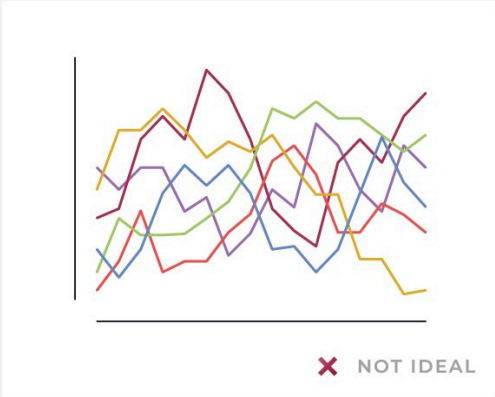


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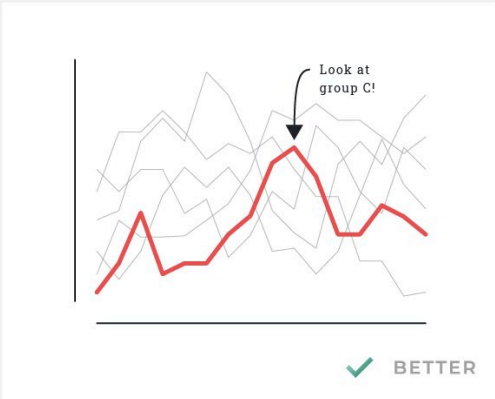
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53 OTHER

THE SPAGHETTI CHART



Too many overlapping lines create a cluttered, unreadable chart. Consider two alternatives: use small multiples for a clearer breakdown or highlight key lines to emphasize the most important groups.

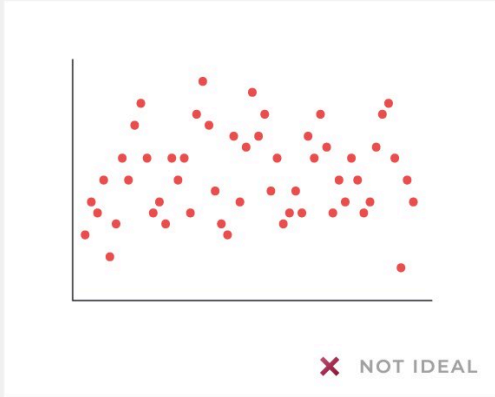


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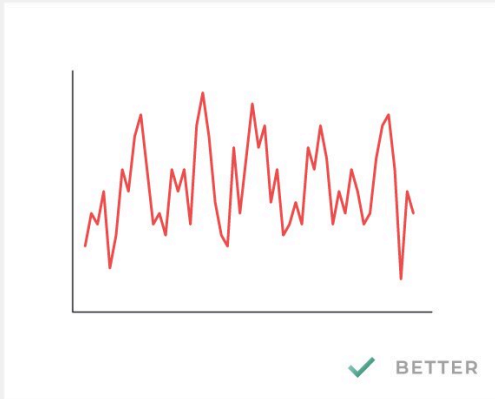
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54 OTHER

CONNECT THE DOTS: REVEAL THE PATTERN



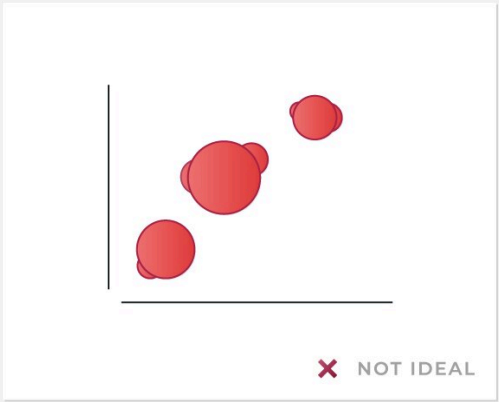
When X-axis points follow a natural order (like time), connecting them with lines makes trends easier to spot.



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Z-INDEX MATTERS



NOT IDEAL

When creating a bubble chart, larger bubbles should be positioned beneath smaller ones, not on top.

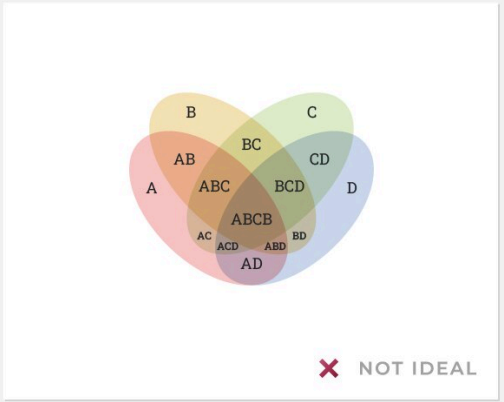


BETTER

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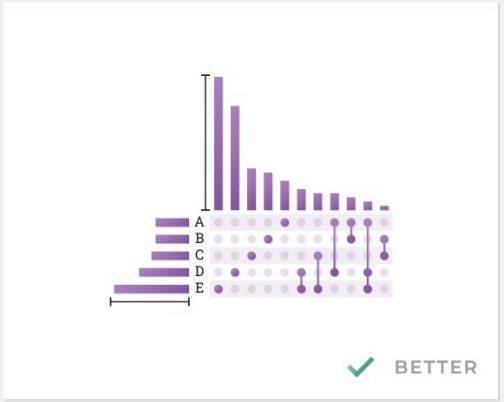
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UPSET PLOT



NOT IDEAL

Venn diagrams quickly become cluttered with 4 or more groups—UpSet plots offer a clearer, more structured way to visualize complex intersections.

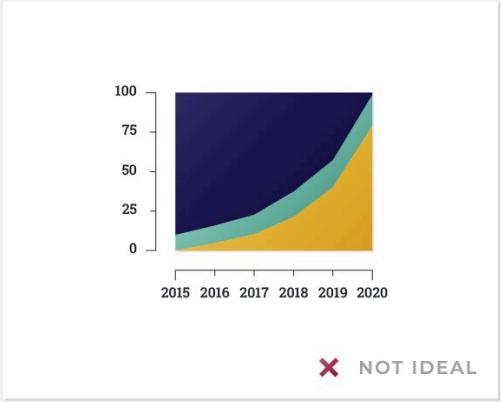


BETTER

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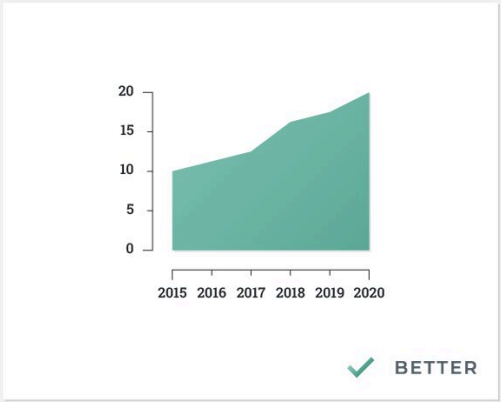
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STACKED CHARTS ARE HARD TO READ



NOT IDEAL

Stacked charts make it hard to track the evolution of a specific group due to the shifting baseline. Consider line charts or small multiples.

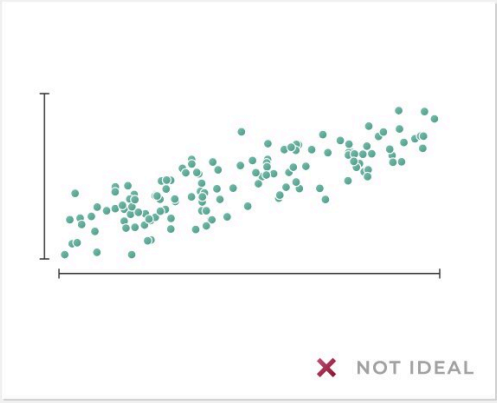


BETTER

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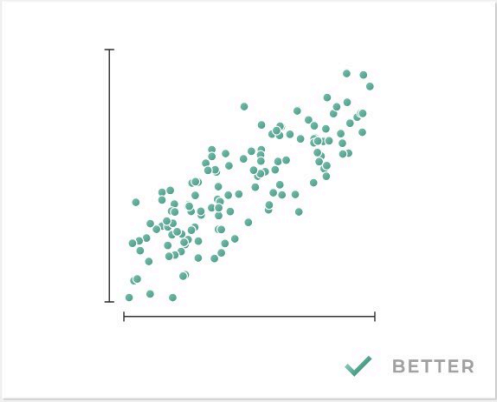
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MIND YOUR SCATTER PLOT'S SHAPE



✗ NOT IDEAL

Stretching a scatter plot changes how steep and how strong the relationship looks. Default to a square plot, and always use a 1:1 ratio when both axes share the same unit.



✓ BETWEEN

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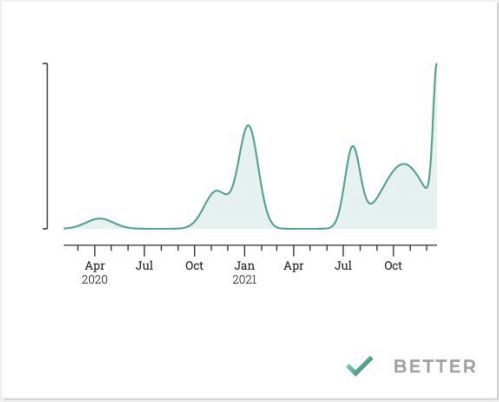
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MAKE DATE AXES EASY TO FOLLOW



✗ NOT IDEAL

Ticks on arbitrary dates, with no year, force readers to work out where they are in time. Place ticks at regular, meaningful intervals such as months or quarters, and show the year.

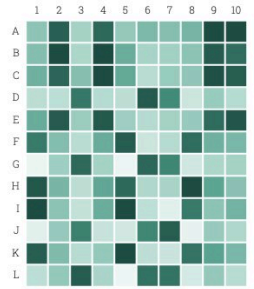


✓ BETTER

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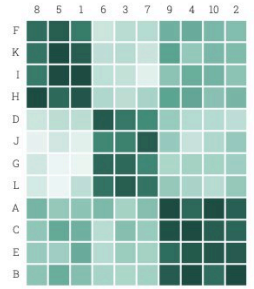
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REORDER YOUR HEATMAP



✗ NOT IDEAL

In alphabetical or random order, the rows and columns of a heatmap look like noise. Reorder them, for instance with hierarchical clustering, so that groups and patterns appear.

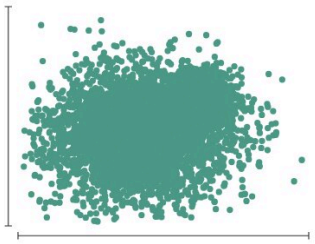


✓ BETTER

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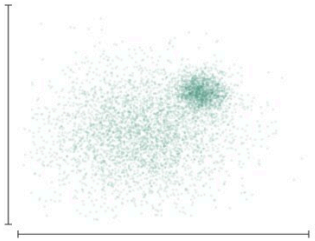
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BEWARE OF OVERPLOTTING



✗ NOT IDEAL

With thousands of points, dots pile up into a solid blob that hides the shape of the data. Use smaller, transparent dots, or switch to a 2D density or hexbin chart.



✓ BETTER

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